

Sean Vincent Confoy

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EDUCATION

Vanderbilt University

Bachelor of Engineering in Mechanical Engineering, Minors in Innovation and Design Strategy | Digital Fabrication

Nashville, Tennessee

August 2022 - May 2026

• **GPA: 3.99/4.0** | Dean's List (4/4) | **SAT:** 1580

Relevant Coursework: Energetics | Mechanics of Materials | Linear Algebra | Aerodynamics | Circuits | Heat Transfer

Davidson Day School

High School Graduate

Cornelius, North Carolina

Graduated May 2022

Weighted GPA: 4.59 **Unweighted:** 4.0/4.0

Honors: Valedictorian | Harvard Book Award | Best Mathematician (2x) | Best Scientist (1x)

DESIGN ENGINEERING SKILLS

CAD/Tools: SolidWorks (CSWA), Fusion 360, Inventor, AutoCAD, Python, MATLAB, LabVIEW, Mathematica

Manufacturing: CNC Mill, Lathe, Welding, Soldering, Composite Manufacturing, Vacuum Forming, Bandsaw

PROFESSIONAL EXPERIENCE

Vanderbilt Aerospace Design Laboratory

Lead Payload Engineer

Nashville, Tennessee

August 2025 – Present

- Engineered retractable shovel payload with multi-stage gearbox, enabling reliable soil extraction for NASA competition rocket
- Led a team of 4 engineers through design reviews, testing, and risk analysis, ensuring deadlines were met under requirements
- Authored technical reports documenting failure analysis and design trade studies presented to faculty and industry reviewers

PAE Engineers

Mechanical Engineering Intern

New York, New York

May 2025 – August 2025

- Engineered HVAC duct layouts meeting ASHRAE airflow standards for high-profile technology clients like Amazon and Intuit
- Revised HVAC and plumbing systems based on client feedback, producing updated schematics and water riser diagrams
- Performed load calculations to determine optimal CFM and evaluated exhaust duct runs to determine static pressure losses

Huntington Ingalls Industries

Mechanical Engineering Intern

Newport News, Virginia

June 2024 – August 2024

- Developed 3D CAD models of naval infrastructure, including a 1.12 million square foot dry dock for housing aircraft carriers
- Produced numerous two-dimensional marine and naval drawing schematics used for renovating foundation and piping systems
- Conducted inspections aboard the USS John F. Kennedy aircraft carrier, evaluating mechanical, electrical, and piping systems

LEADERSHIP & PROJECTS

Tikkun Olam Makers

Mechanical Engineer

Nashville, Tennessee

September 2024 – Present

- Analyzed needs of patient with severe condition similar to cerebral palsy and conceptualized potential projects to assist them
- Formulated multiple designs to address patients' difficulty to hold everyday objects and inability to avoid looking toward sunlight
- Combined brainstormed solutions into working prototypes which vastly streamlined patients' ability to enjoy simple pleasures

Vanderbilt Fusion Project

Fusion Researcher

Nashville, Tennessee

February 2024 – February 2025

- Designed CAD models for a pelican case and control panel to house buttons, switches, and wiring for fusion reactor startup
- Fabricated a motor-driven, switch-operated, and partially 3d printed mechanism for supplying power to the fusion reactor

MEMBERSHIPS AND INTERESTS

Security Clearance: U.S. Secret (in process; initiated via HII internship)

Memberships: Tau Beta Pi Honor Society | Sigma Nu

Interests: Rock Climbing | Jiu Jitsu | Scuba Diving | Exercise | Archaeology | Snowboarding